



SOP – Filament 3D printer

Do not operate the machine if you haven't received the appropriate technical instruction.

You can only be instructed by the machine-responsible persons:

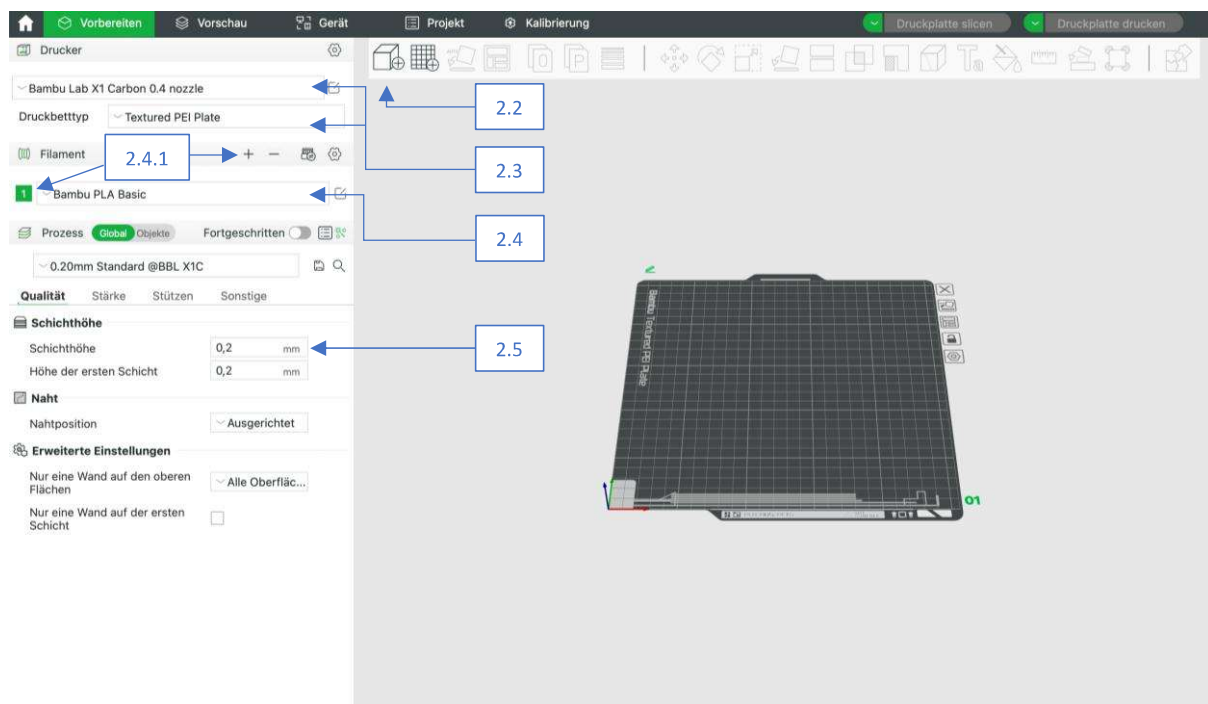
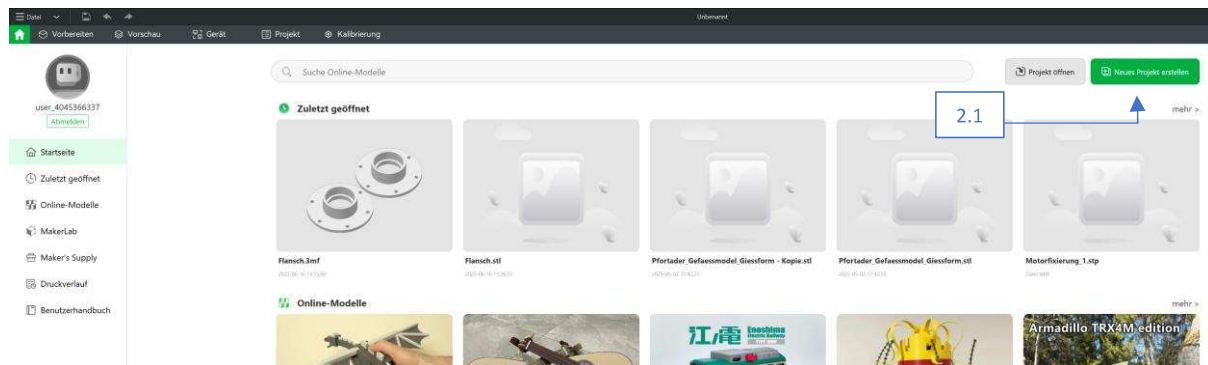
Tobias Mörschel, M. Sc.

I. Preparation for 3D-printing (Slicing) via Bambu Studio

1. Requirements

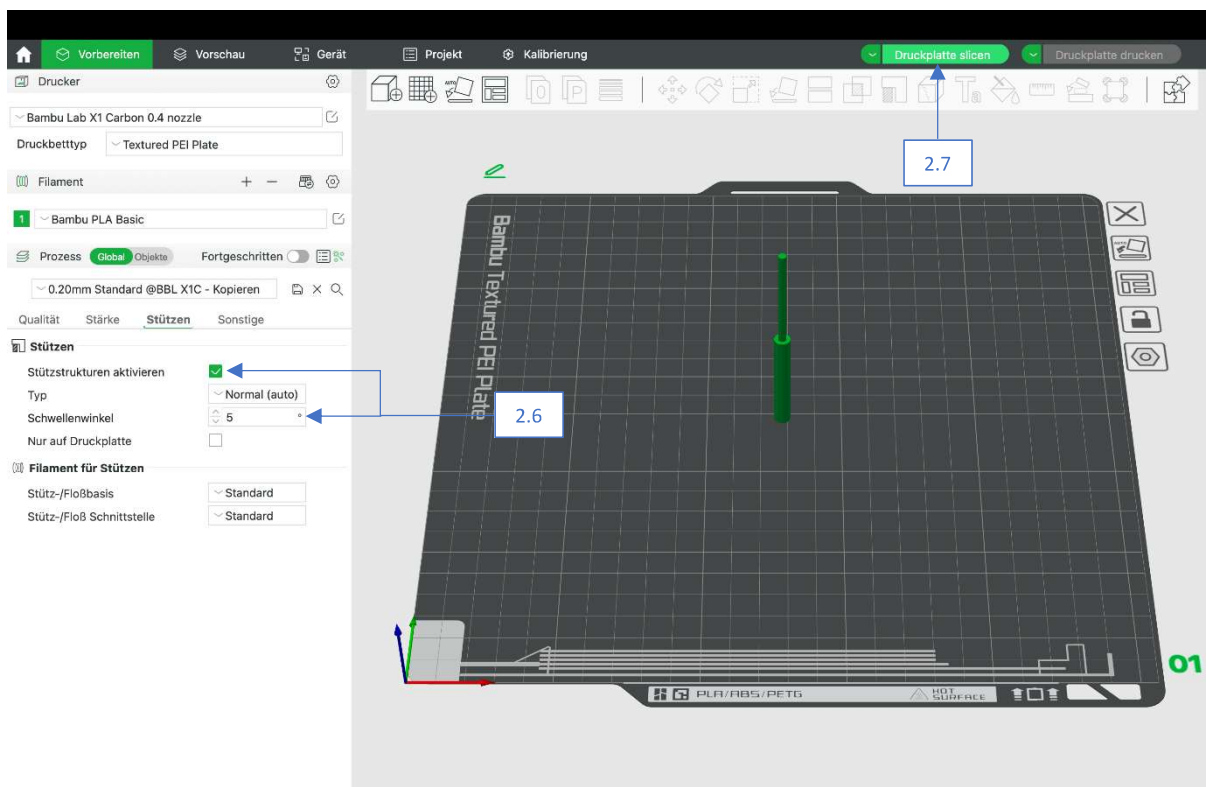
- 1.1 Bambu Studio installed <https://bambulab.com/de-de/download/studio>
- 1.2 The saved STEP- or STL-file of part from CAD software (e.g. Siemens NX)
- 1.3 Use the designated SD card to transfer the sliced file between the computer and the 3D printer. **The SD card and USB stick should remain with the printer and only be used to export ready files.**

2. Loading the part into Bambu Studio to prepare it for the 3D print



- 2.1 Open a new project by clicking 'Neues Projekt erstellen'.

- 2.2 Click on the cube icon to import a model. Alternatively drag and drop the model files into the program.
- 2.3 Within the tab 'Vorbereiten', choose the Printer settings under 'Drucker'
 - 2.3.1 Select Printer 'Bambu Lab X1 Carbon 0.4 nozzle'
 - 2.3.2 Select Druckbetttyp 'Textured PEI Plate'
- 2.4 Choose the exact filament you will be using (e.g. PLA white, PLA green, PC, PLA for Support structures) under 'Filament' from the Dropdown-Menu. If you intend to use the PLA for Support structures, please use as little as possible.
 - 2.4.1 Choose the color on the number to the left. The color only has to be similar to the actual filament. Add multiple colors by the plus sign.
- 2.5 Choose the layer height under 'Qualität' option 'Schichthöhe'. Remember that the smaller layer height, the longer the printing takes. Recommended layer heights are between 20% and 70% of the nozzle diameter, for the 0.4 mm nozzle the standard layer height is 0.2 mm.



- 2.6 Make sure the box 'Stützstrukturen aktivieren' is checked if necessary. Set 'Schwellenwinkel' to 5°. This defines the degree of overhang for which support structures are calculated. Check for critical overhangs with the tool shown in 2.7.2. and increase the 'Schwellenwinkel' if necessary.



Druckplatte drucken

Linientyp	Zeit	Prozent	Genutztes Filament
Innere Wand	3m23s	16,6%	0,13 m 0,41 g
Außenwand	4m39s	22,9%	0,16 m 0,47 g
Füllung	2m34s	12,6%	0,09 m 0,28 g
Innere massive Füllung	1m0s	4,9%	0,03 m 0,08 g
Oberfläche	2s	0,2%	0,00 m 0,01 g
Untere Fläche	1s	0,1%	0,00 m 0,00 g
Lückenfüllung	<1s	<0,1%	0,00 m 0,00 g
Umrandung	12s	1,0%	0,02 m 0,05 g
Benutzerdefiniert	6m57s	34,2%	0,11 m 0,33 g
Eitgang	1m46s	8,7%	
Rückzug			
Einzug			
Reinigen			
Nähte			

Gesamtschätzung

Total Filament: 0,54 m 1,64 g
Model Filament: 0,54 m 1,64 g
Kosten: 0,04
Vorbereitungszeit: 6m42s
Druckzeit des Modells: 13m40s
Gesamtdauer: 20m22s

Progress bar: 110 / 220 (1 / 0,20)

Exportieren der Datei einer geslicten Druckplatte

Filament	Modell
1	0,54 m 1,64 g

Filamentwechselzeiten: 0
Kosten: 0,04

Geschätzte Zeit

Vorbereitungszeit: 6m42s
Druckzeit des Modells: 13m40s
Gesamtdauer: 20m22s

Optionen

Eitgang: ☐
Rückzug: ☐
Einzug: ☐
Reinigen: ☐
Nähte: ☒

Progress bar: 400 / 0,00 (1 / 0,20)



- 2.7 To slice, click the button 'Druckplatte slicen'.
 - 2.7.1 The box 'Farbschema' will appear, allowing you to check the estimated printing duration. Please be considerate of others who want to use the Printer as well.
 - 2.7.2 By moving the axis on the right, you can check how the specific layers will look, and what line type they will be. If an error occurs while slicing, slice again after fixing the error.
- 2.8 Choose 'Filament' from the dropdown menu. If you use multiple filaments, discuss it first with Tobias Mörschel. Then position the parts in such a way to reduce filament changes (counted under 'Filamentwechselzeiten') due to filament changes increasing material consumption and printing duration.
- 2.9 Choose 'Exportieren der Datei einer gesliceten Druckplatte' from the dropdown menu and then click on it to save it onto the SD Card. Eject the SD card before removing it from your PC to avoid file corruption.

II. Printing

No one can start a print without permission

Students should be present before the print starts. If you notice an error or something not working let Tobias Mörschel know.

Before starting a new print, verify that no remnants from the previous print remain on the bed or the nozzle, particularly calibration material.

1.4 Start print by using SD Card. Remember: **The SD card and USB stick should remain with the printer and only be used to export ready files.**



- 1.4.1 Insert the SD-Card into the right side of the display. Press the folder icon and select the tab 'SD Card'
- 1.4.2 Check if the filament you chose is available.
- 1.4.3 Choose the sliced file you want to print. Check the filament settings.
- 1.4.4 Make sure the printers' door is closed and no remaining filament is left in the printer.
- 1.4.5 Start the printing process by pressing 'Print'. The program will give you a rough estimation, when the print will be done.

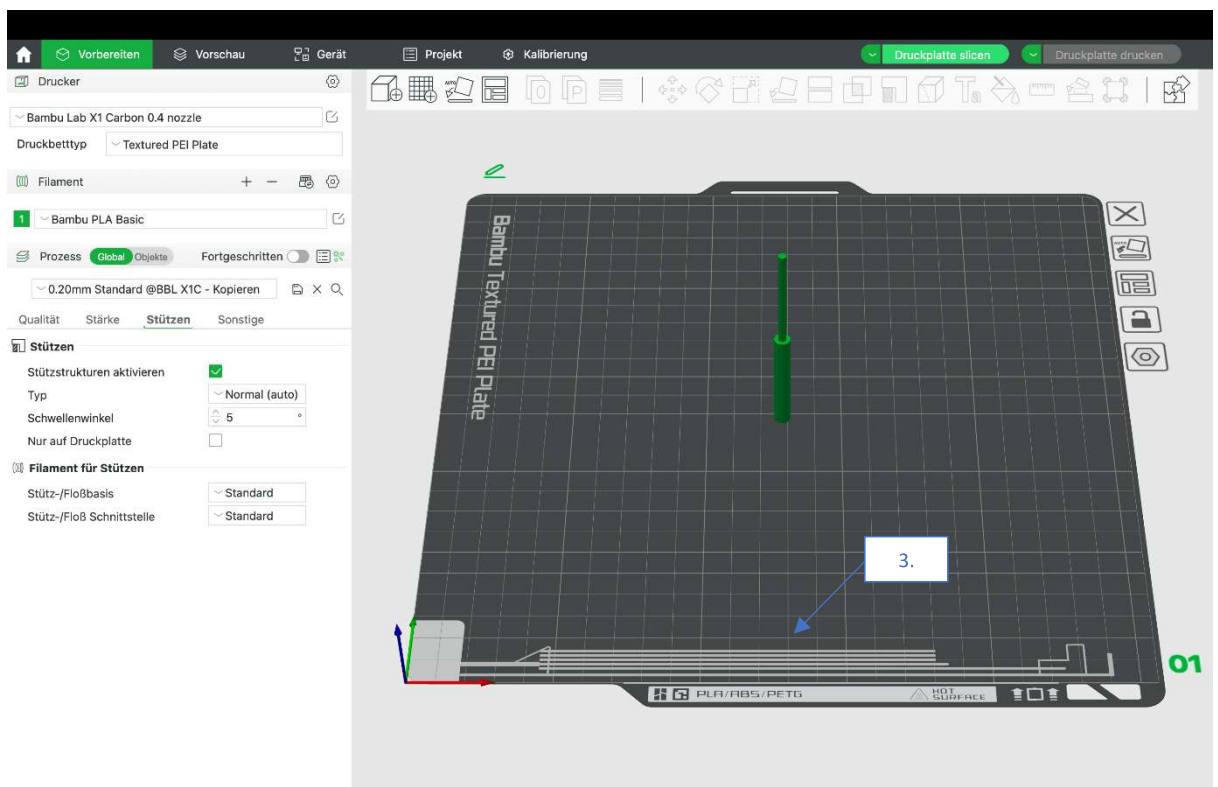
Never reach into a running print. Be careful with long hair, chains or necklaces when looking into the 3D printer from above

1.5 During printing

- 1.5.1 If filament strings form, pause the process with 'Pause' and only open the door once the printing head has moved to the back wall. Wait for the printer to stop moving, before reaching in. Remove the filament strings and press 'Resume'. **If you see something going obviously wrong, even if it is not your print, press 'Pause' and tell Tobias Mörschel.**
- 1.5.2 If you wish to cancel your own print, press 'Stop'. Only open the door when the printing head has moved to the back wall. Wait for the printer to stop moving, before reaching in.

III. After the print

The print is finished when the display says 100%. Always wait for this before opening the door, because the printer might move again.



1. Remove parts using the bed scraper.
2. **Some parts could be difficult to detach from the platform. Please be careful not to injure yourself or the printer while removing the parts or wear safety gloves.**
3. Remove all other filament remnants from the printing bed, don't forget those from calibration.
4. Remove support structures and extra parts with the tools in the workstation. Always wear safety glasses!



5. Use sandpaper to smooth surfaces if needed.

IV. Leaving the working space

1. Clean and return all used tools to their proper places. Ensure that the workspace is tidy for the next user.
2. Dispose of the used filament by putting it into the garbage can below the printer.